Intro & Data

Traffic stops are among the most common interactions between civilians and police officers in the United States. During these encounters, officers exercise substantial discretion in deciding whether to issue warnings, conduct searches, make arrests, or escalate the stop further. Because of this discretion, researchers and the public have increasingly questioned whether demographic characteristics such as race and gender influence traffic stop outcomes.

This project investigates whether race and gender are associated with differences in police stop outcomes in Massachusetts. Specifically, the analysis examines disparities in searches, arrests, contraband findings, and stop escalation across demographic groups. The project was motivated by prior research, including the Stanford Open Policing Project study by Emma Pierson and collaborators, which identified racial disparities in police stop and search practices across the United States.

The analysis uses Massachusetts traffic stop data from the Stanford Open Policing Project covering December 2006 through December 2015. Variables selected for the study include driver race, driver gender, whether a search was conducted, whether contraband was found, and whether an arrest occurred. Race and gender serve as explanatory variables, while search outcomes, arrests, and escalation measures serve as response variables.

Using descriptive statistics, confidence intervals, logistic regression, and a bias index analysis, this project evaluates whether police stop outcomes differ systematically across demographic groups. The group hypothesized that minority drivers and male drivers would experience higher rates of searches and arrests and that disparities would become more pronounced when race and gender were considered jointly.

Descriptive Analysis

To examine how race and gender relate to police stop outcomes in Massachusetts, the group conducted an exploratory data analysis using summary statistics and visualizations. The analysis focused on both the distribution of traffic stops and the likelihood of escalated outcomes such as searches and arrests.

Figures 1 and 2 show the distribution of traffic stops across race and gender. White drivers accounted for the majority of recorded traffic stops, while minority groups represented a smaller share of the dataset. Similarly, male drivers made up a substantially larger proportion of stops than female drivers. However, these distributions may partially reflect the demographic composition and driving patterns of Massachusetts rather than differences in policing practices alone.

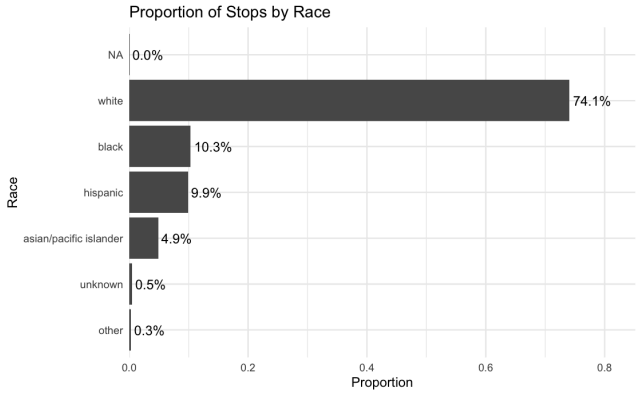


Figure 1: Proportion of traffic stops by race in Massachusetts.

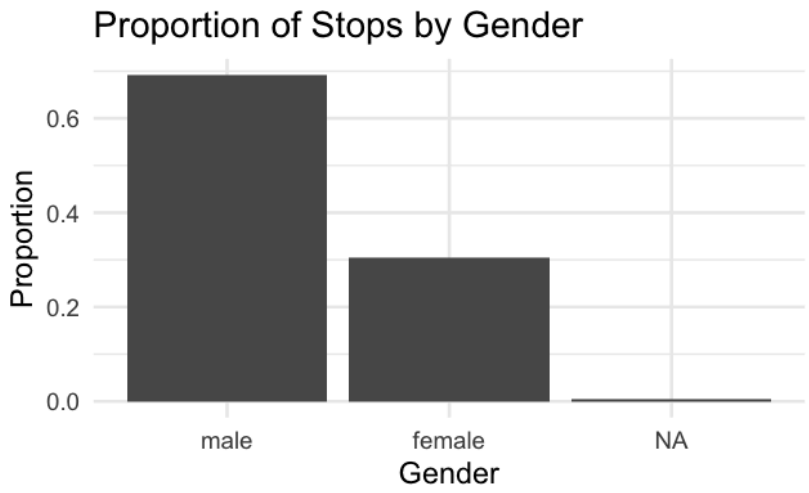


Figure 2: Proportion of traffic stops by gender in Massachusetts.

The group then examined overall stop outcomes. Figure 3 shows that escalated outcomes were relatively uncommon across all traffic stops. Approximately 1.6% of stops resulted in a search, about 0.8% resulted in contraband being found, and roughly 2.7% resulted in an arrest. Although these events were rare overall, differences across demographic groups remained important to investigate.

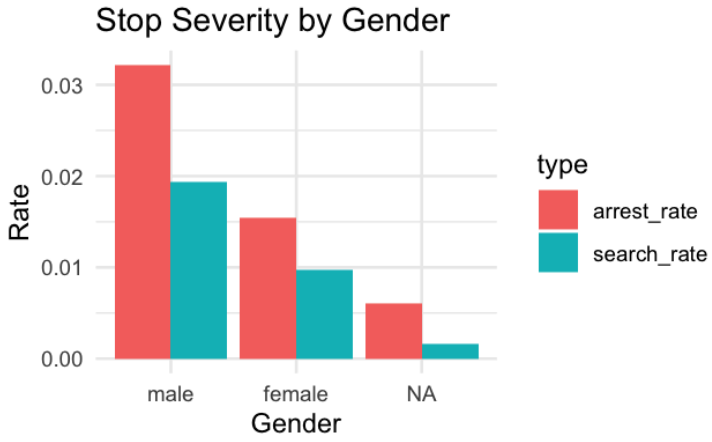


Figure 3: Overall probabilities of search, contraband found, and arrest during traffic stops.

Figures 4 and 5 compare search and arrest rates across gender and race. Male drivers experienced substantially higher search and arrest rates than female drivers. Across racial groups,

Hispanic drivers showed the highest search and arrest rates, followed by Black drivers, while White drivers consistently exhibited the lowest rates. These patterns suggest an association between demographic characteristics and the likelihood of escalated police stop outcomes.

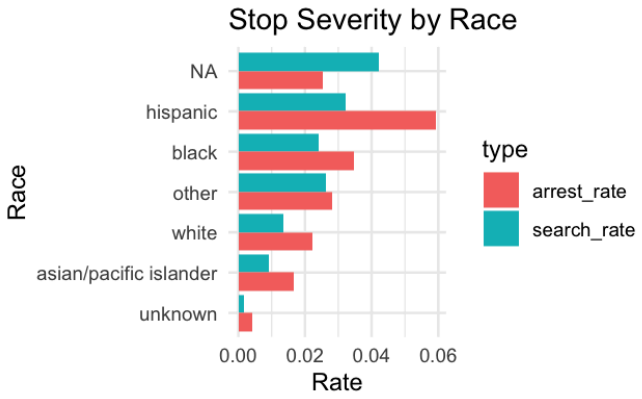


Figure 4: Search and arrest rates by gender in Massachusetts traffic stops.

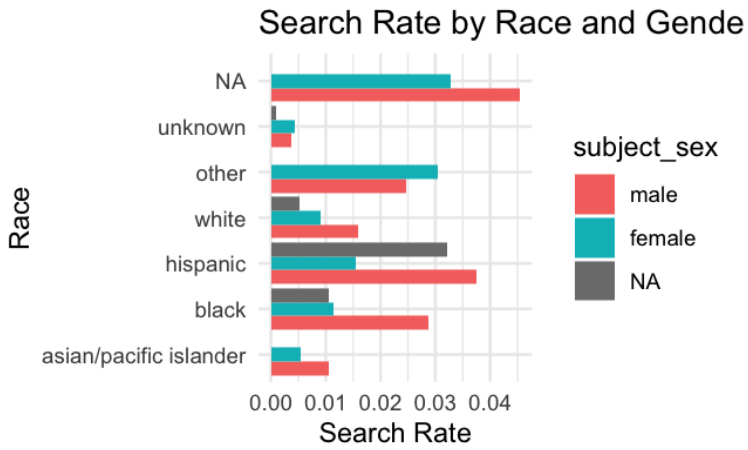


Figure 5: Search and arrest rates by race in Massachusetts traffic stops

To further examine these relationships, the group analyzed outcomes across combined race–gender categories. Figure 6 suggests that disparities become more pronounced when these variables are considered jointly. Minority males—particularly Black and Hispanic males—experience the highest rates of searches and arrests among all groups. This suggests an

interaction effect between race and gender, where the combined impact is greater than either variable alone.

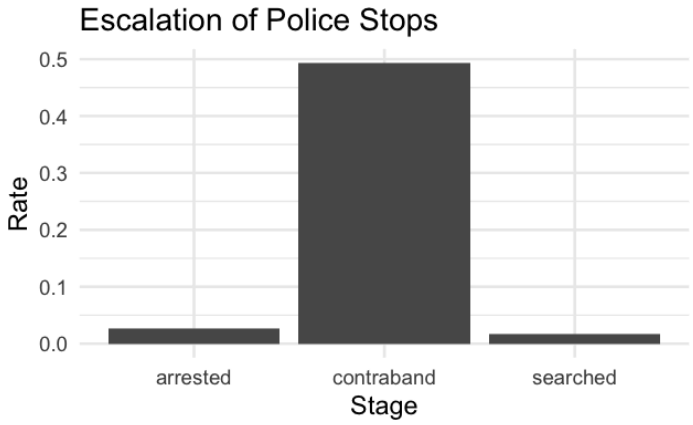


Figure 6: Search and arrest rates across race–gender groups.

Overall, the exploratory analysis identified several consistent patterns. While White and male drivers accounted for the largest share of total stops, minority drivers and male drivers experienced higher rates of searches and arrests once stopped. Additionally, the combination of race and gender appeared to amplify these disparities, particularly among Black and Hispanic males.

Because these findings are based on descriptive statistics and visual trends, they do not establish causation or statistical significance. Therefore, inferential methods such as confidence intervals and logistic regression were used to further evaluate whether the observed differences were statistically meaningful.

Inference

To analyze the first two graphs representing the proportion of stops by race and gender, we created two confidence intervals at 95% confidence level. For the confidence interval analyzing race, the goal was to examine whether white drivers (white count) in Massachusetts

were stopped more frequently than drivers of other races (nonwhite count). Based on the calculated proportions, about 74.1% of traffic stops involved white drivers and the other 25.9% of traffic stops involved drivers who were not white. Accordingly, we are 95% confident that the true difference between the proportion of stops with white drivers and nonwhite drivers is between 0.481 and 0.482. Since the confidence interval is above 0, white drivers make up a large proportion of stops in the data set.

```
        2-sample test for equality of proportions without continuity correction

data:  c(white_count, nonwhite_count) out of c(total_race, total_race)
X-squared = 1584958, df = 1, p-value < 2.2e-16
alternative hypothesis: two.sided
95 percent confidence interval:
 0.4810972 0.4824116
sample estimates:
   prop 1    prop 2
0.7408772 0.2591228
```

Figure 7: Confidence interval for the proportion of stops by race.

For the confidence interval for gender, the goal was to examine whether male drivers (male count) in Massachusetts were stopped more frequently than female drivers (female count). From the calculated proportions, about 69% of traffic stops involved male drivers and the other 31% involved female drivers. Thus, we are 95% confident that the true difference between the proportion of stops with male drivers and female drivers is between 0.388 and 0.389. Because the confidence interval is entirely above 0, the confidence interval shows that male drivers represent a larger proportion of the data set.

```
        2-sample test for equality of proportions without continuity correction

data:  c(male_count, female_count) out of c(total_male_female, total_male_female)
X-squared = 1030759, df = 1, p-value < 2.2e-16
alternative hypothesis: two.sided
95 percent confidence interval:
 0.3886082 0.3899927
sample estimates:
   prop 1    prop 2
0.6946502 0.3053498
```

Figure 8: Confidence interval for the proportion of stops by gender.

Next in analyzing the search and arrest rates along the lines of race and gender, a confidence interval was made with each variable. For the gender confidence interval, the results were displayed in a table where the search_rate column represents the proportion of stops that involved a search and the arrest_rate column represents the proportion of stops that involved an arrest. The female search rate shows that close to 1% of stops lead to searches. For males, that percentage was close to 2%. The male search rate is about 2 times higher than the female search rate. Because the intervals do not overlap, there is strong evidence to suggest that the difference is legitimate. In terms of arrest rates, the female percentage was around 1.5% while the male percentage was around 3.15%. Similarly, the male arrest rate is about 2 times higher than the female arrest rate and the lack of overlap suggests the difference is real.

A tibble: 3 × 8

subject_sex <chr>	n <int>	search_success <int>	arrest_success <int>	search_low <dbl>	search_high <dbl>	arrest_low <dbl>	arrest_high <dbl>
female	1038377	10036	16005	0.009478236	0.009855568	0.015177840	0.015652706
male	2362238	45668	75920	0.019157497	0.019509097	0.031914653	0.032364898
NA	15623	26	94	0.001110057	0.002475423	0.004890829	0.007391955

3 rows

Figure 9: Confidence interval for search and arrest rates with relation to gender.

For the race confidence interval, the arrest rates for white, black, and hispanic drivers were about 2.2%, 3.4%, and 5.9%, respectively. The data show that hispanic drivers face the highest arrest

rates while white drivers face the lowest arrest rates. There is little to no overlap so the differences are likely true.

A tibble: 7 × 8

subject_race <chr>	n <int>	search_success <int>	arrest_success <int>	search_low <dbl>	search_high <dbl>	arrest_low <dbl>
asian/pacific islander	166842	1539	2748	0.008773817	0.009697522	0.01586805
black	351610	8428	12225	0.023467939	0.024481970	0.03416677
hispanic	338317	10944	20085	0.031756012	0.032951353	0.05857463
other	11008	288	310	0.023299334	0.029361869	0.02518869
unknown	17017	26	69	0.001019105	0.002272728	0.00317953
white	2529780	34435	56540	0.013469609	0.013755582	0.02216814
NA	1664	70	42	0.033163294	0.053146042	0.01847052

7 rows | 1–7 of 8 columns

Figure 10: Confidence interval for search and arrest rates with relation to race.

For the next portion of inference in regards to the search rate based on both race and gender, a logistic model with an interaction model was created. Males have an increased 94% chance of being searched compared to female drivers. When sex was controlled for, black and hispanic drivers were more likely to be searched. With race as the baseline, male drivers still have double the chance of being searched. If an individual was both black and male, the severity of the stop was increased and this also happened for hispanic males. Interestingly for other races, being a male driver actually reduced the severity of the traffic stop. Since the p-value was 0.15, however, there is no strong evidence that gender impacts the chances of being searched as a white Massachusetts driver.

```
Call:
glm(formula = search_conducted ~ subject_race + subject_sex,
    family = binomial, data = massachusetts_data)

Coefficients:
                    Estimate Std. Error  z value Pr(>|z|)
(Intercept)         -5.21012    0.02732 -190.691  < 2e-16 ***
subject_raceblack    0.98559    0.02789   35.336  < 2e-16 ***
subject_racehispanic 1.27101    0.02740   46.387  < 2e-16 ***
subject_raceother    1.12184    0.06504   17.248  < 2e-16 ***
subject_raceunknown -0.79538    0.25935   -3.067  0.00216 **
subject_racewhite    0.43698    0.02619   16.684  < 2e-16 ***
subject_sexmale      0.66166    0.01113   59.463  < 2e-16 ***
---
Signif. codes:  0 ‘***’ 0.001 ‘**’ 0.01 ‘*’ 0.05 ‘.’ 0.1 ‘ ’ 1

(Dispersion parameter for binomial family taken to be 1)

    Null deviance: 567934  on 3398954  degrees of freedom
Residual deviance: 556629  on 3398948  degrees of freedom
  (17283 observations deleted due to missingness)
AIC: 556643

Number of Fisher Scoring iterations: 7


Call:
glm(formula = search_conducted ~ subject_race * subject_sex,
    family = binomial, data = massachusetts_data)

Coefficients:
                                Estimate Std. Error z value Pr(>|z|)
(Intercept)                     -5.22014    0.06741 -77.435  < 2e-16 ***
subject_raceblack                0.75901    0.07393  10.267  < 2e-16 ***
subject_racehispanic             1.06527    0.07337  14.520  < 2e-16 ***
subject_raceother                1.75933    0.11872  14.819  < 2e-16 ***
subject_raceunknown             -0.21649    0.41403  -0.523  0.60106
subject_racewhite                0.52155    0.06842   7.623 2.48e-14 ***
subject_sexmale                  0.67337    0.07288   9.240  < 2e-16 ***
subject_raceblack:subject_sexmale     0.26631    0.07983   3.336  0.00085 ***
subject_racehispanic:subject_sexmale  0.23506    0.07910   2.972  0.00296 **
subject_raceother:subject_sexmale    -0.88914    0.14338  -6.201 5.60e-10 ***
subject_raceunknown:subject_sexmale  -0.83515    0.53201  -1.570  0.11646
subject_racewhite:subject_sexmale    -0.10583    0.07407  -1.429  0.15306
---
Signif. codes:  0 ‘***’ 0.001 ‘**’ 0.01 ‘*’ 0.05 ‘.’ 0.1 ‘ ’ 1

(Dispersion parameter for binomial family taken to be 1)

    Null deviance: 567934  on 3398954  degrees of freedom
Residual deviance: 556384  on 3398943  degrees of freedom
  (17283 observations deleted due to missingness)
AIC: 556408

Number of Fisher Scoring iterations: 7
```

Figures 11 and 12: Logistic model and interaction model for search rate by race and gender.

Finally, to analyze the escalation of police stops, one more confidence interval was made at 95% confidence level. Because the sample size was very large (3,416,238), the intervals were tight. We are 95% confident that the true probability of a stop leading to a search is between

1.62% to 1.64%. In the contraband stage, we are 95% confident that the true probability of contraband being found is between 0.795% to 0.814%. At the arrest stage, we are 95% confident that the true probability of being arrested is between 2.676% and 2.711%.

n <int>	searched_success <int>	contraband_success <int>	arrested_success <int>	search_rate <dbl>	contraband_rate <dbl>
3416238	55730	27474	92019	0.01631327	0.00804218

1 row | 1-6 of 13 columns

arrest_rate <dbl>	search_low <dbl>	search_high <dbl>	contraband_low <dbl>	contraband_high <dbl>	arrest_low <dbl>	arrest_high <dbl>
0.02693577	0.01617933	0.01644829	0.007947873	0.008137594	0.02676448	0.02710813

1 row | 7-13 of 13 columns

Figure 13: Confidence interval for the escalation of police stops.

Extended Analysis: Bias Index

To extend the analysis beyond basic search and arrest comparisons, the group created a bias index to examine search efficiency across demographic groups. This index compares how often a group is searched to how often those searches result in contraband being found. The bias index was calculated as: Bias Index = Search Rate ÷ Hit Rate, where search rate is the proportion of stops that resulted in a search, and hit rate is the proportion of searches that resulted in contraband being found. A higher bias index suggests that a group is searched frequently relative to how often contraband is actually discovered.

This measure adds depth to the analysis because it does not only show which groups are searched more often; it also evaluates whether those searches are equally effective. If a group has a high search rate but a lower hit rate, this may suggest that increased search frequency is not fully supported by higher contraband discovery. In this way, the bias index helps connect the statistical findings more directly to questions of fairness and consistency in police discretion.

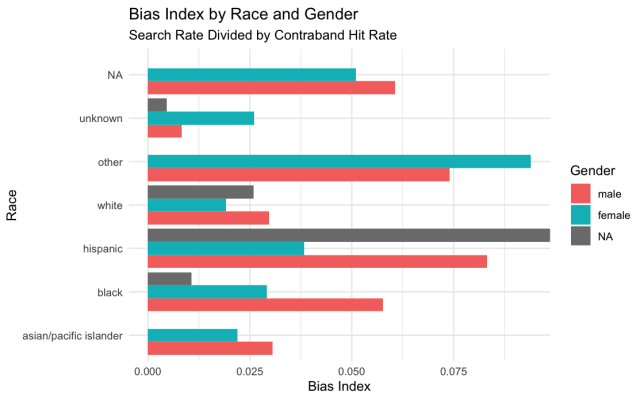


Figure 14: Bias index by race and gender in Massachusetts traffic stops.

Overall, the bias index provides an additional way to interpret disparities in traffic stop escalation. While earlier analyses showed that race and gender were associated with different search and arrest rates, this extended analysis considers whether those searches were productive at similar rates across groups. This strengthens the project by moving beyond identifying disparities and toward evaluating the efficiency of those disparities

Discussion

The purpose of this paper was to analyze if someone's race or gender had an impact on the outcome of a traffic stop in Massachusetts. That claim was supported: the statistical analyses conducted suggest that demographic characteristics are related to stop severity, searches, arrests, and escalation during traffic stops.

The exploratory data analysis revealed several consistent patterns: White drivers and male drivers accounted for the largest proportion of total traffic stops; however, minority drivers—particularly Hispanic and Black drivers—experienced higher search and arrest rates once stopped. Similarly, males were searched and arrested at much higher rates compared to females. When race and gender were analyzed together, Black and Hispanic males experienced the highest levels of stop severity among all demographic groups. It is important to note here that

White drivers were likely stopped the most because about 78% of the population of Massachusetts is White (US Census Bureau).

The confidence intervals that were previously performed showed meaningful differences in search and arrest rates across demographic groups, while the logistic regression and interaction models indicated that both race and gender are associated with increased search likelihood. The extended bias index analysis further suggested that some demographic groups experience higher search rates without proportionally higher contraband discovery rates, indicating potential differences in search efficiency across groups. Although searches, contraband findings, and arrests are uncommon overall, their unequal distribution across demographics suggests that someone's race and gender may influence traffic stop outcomes; however, because this analysis is observational, the results do not establish direct causation.

Future research should examine the reason for each traffic stop, compare data across different areas based on population density and socio-economic status, and should take into account the policies and "cultures" of individual police stations to provide greater context for understanding the disparities observed in this analysis. It is also crucial to highlight that there is no data on who violates traffic laws without getting pulled over, which may be noteworthy in further exploring the issue of biased profiling.

References

Pierson, Emma, et al. “Data - The Stanford Open Policing Project.” *stanford.edu*, 2020, https://openpolicing.stanford.edu/data/. Accessed 12 May 2026.

Pierson, Emma, et al. “A large-scale analysis of racial disparities in police stops across the United States.” *stanford.edu*, 2020, 5harad.com/papers/100M-stops.pdf. Accessed 12 May 2026.

“US Census Bureau QuickFacts: Massachusetts.” *United States Census Bureau*, 1 July 2025, census.gov/quickfacts/fact/table/MA/PST045225. Accessed 12 May 2026.

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